

Supporting Active Learning in Math Tasks Through a RAG-based Gamified System

Michail PETTAS ^a, ZhiJun CHENG ^a and Joris LEIPUS ^a

^a*Master's Programme in Artificial Intelligence, 2025*

Abstract. Mathematics classrooms often struggle to support students with widely different skill levels, leading to disengagement for some and anxiety for others. Standard AI chatbots can worsen this by providing direct answers, bypassing the reasoning process that learning depends on. This project presents the design of a gamified web-based tutoring system for primary school students aged 10–12, where mathematical challenges are embedded in a turn-based card combat game. The system uses a Retrieval-Augmented Generation (RAG) architecture constrained by the “MiniGuide” pedagogical strategy, designed so that the AI provides guiding questions rather than direct answers, functioning as a supportive peer that aims to keep learners within their Zone of Proximal Development (ZPD). An exploratory evaluation with a teacher and students evaluates whether the interaction design produces appropriate pedagogical guidance and an engaging learning experience.

Keywords. Retrieval-Augmented Generation, Active Learning, User Modeling, Gamification, Mathematics Education, Participatory Design,

1. Introduction

Mathematics classrooms face a persistent challenge: a typical class of 10 to 12 year-olds often contains students ranging from a primary grade level to a lower secondary level, creating a situation where high performers frequently become bored, while low performers experience anxiety and helplessness [1]. Traditional teaching methods often struggle to address these individual needs simultaneously.

The recent emergence of Large Language Models (LLMs) offers new possibilities for personalized tutoring, and as such, research and implementation have shifted from “one-size-fits-all” methods towards AI-driven personalized approaches. Models such as Personalized Adaptive Gamified e-Learning (PAGE) use machine learning to process student data and dynamically adjust game mechanics, content difficulty, and feedback based on real-time progress, delivering customized learning paths aligned with individual cognitive profiles [2]. At the modeling level, Reinforcement Learning (RL) and user modeling techniques construct representations of a learner’s goals, knowledge, and misconceptions, then optimize difficulty levels and reward structures to keep learners in a productive zone between boredom and frustration [3].

While these tools offer significant potential for personalization, current implementations often present new risks to the learning process [4]. Students frequently utilize AI tools to obtain immediate answers rather than to understand underlying concepts, leading to a superficial “illusion of knowledge” where material is not fully internalized. Further-

more, standard LLMs can suffer from “hallucinations”, producing confident but incorrect mathematical reasoning, which undermines trust in educational settings [5]. To address these issues, educational AI must shift from being an oracle that delivers answers to a “More Knowledgeable Peer” that facilitates Active Learning, encouraging students to internalize material through interactive engagement and constructive problem-solving.

Making this shift concrete requires grounding the system in how learning actually works. Research on cognitive agency suggests that reasoning skills develop best when learners actively engage with problems rather than receive ready-made answers [6,7], and Self-Determination Theory identifies autonomy, competence, and relatedness as the psychological conditions under which this kind of intrinsic engagement is most likely to occur [8]. At the instructional level, the “MiniGuide” strategy provides a structured approach: rather than resolving a student’s difficulty directly, the tutor poses guiding questions that keep the learner doing the cognitive work [9].

2. Aim and Research Questions

The primary aim of this project is to investigate how Active Learning in primary mathematics education (ages 10–12) can be systematically supported through a responsible, interactive AI system. Specifically, the project examines how a Retrieval-Augmented Generation (RAG) architecture can be designed to function as a “More Knowledgeable Peer” [10], providing adaptive scaffolding that aligns with diverse learner profiles. Unlike standard chatbots, a RAG architecture helps improve transparency and factual consistency by grounding AI responses in verified mathematical curricula, thereby mitigating the black-box nature of LLM reasoning [11]. By combining adaptive support driven by a dynamic user model with game-based interaction elements to mitigate anxiety [12], the system seeks to transform mathematics learning into an engaging, low-stakes, and cognitively supportive experience.

To achieve this aim, the project investigates the following research questions:

- RQ1:** How can a dynamic user model be operationalized within a RAG framework to approximate a student’s Zone of Proximal Development and deliver guidance that promotes active reasoning rather than answer dependency?
- RQ2:** To what extent can game-based design reframe mathematical errors as constructive learning moments, thereby reducing anxiety and sustaining motivation?
- RQ3:** How can the “MiniGuide” pedagogical strategy enable the system’s generation logic to ensure that AI responses provide transparent guidance rather than direct answers?

3. Theoretical Grounding

3.1. *Self-Determination Theory*

Self-Determination Theory (SDT), developed by Ryan and Deci, proposes that intrinsic motivation is driven by three basic psychological needs: autonomy, competence, and relatedness [8]. In educational contexts, this implies that learning systems should offer meaningful choices, provide challenges matched to the learner’s ability, and foster a sense of connection or support rather than isolation.

3.2. *Activity Theory*

Activity Theory, rooted in the work of Vygotsky, provides a framework for analyzing human activity as a system involving a subject, an object, and mediating instruments, situated within a community governed by rules and division of labor [10]. It is particularly useful for identifying contradictions within a system, such as when the demands on a role exceed what one person can provide, and for reasoning about how new tools or artifacts might address those contradictions.

3.3. *Active Learning*

Active Learning refers to instructional approaches that require learners to engage in meaningful cognitive activities, such as problem-solving and reflection, rather than passively receiving information [6,7]. In the context of AI-assisted education, this principle suggests that systems should promote reasoning and effort rather than simply delivering answers.

3.4. *MiniGuide*

The MiniGuide strategy, developed at Umeå University, is a pedagogical approach in which a tutor supports a learner's problem-solving by posing guiding questions rather than providing direct answers [9]. It operationalizes the concept of scaffolding within the Zone of Proximal Development, aiming to help learners develop their own reasoning rather than creating dependence on external solutions.

4. **Methods**

We employ a participatory design process defined by iterative cycles of co-design, prototyping, and reflection [13]. Collaborative tasks range from identifying learning goals to refining prototypes with stakeholders, including 10 to 12 year-old learners and educators. Stakeholders contribute domain knowledge, while our team manages prototyping, and ensures theoretical alignment.

4.1. *Analysis of the Use Situation*

The use situation is situated within the mathematics classroom for students aged 10–12. While the subject (the student) aims to acquire mathematical reasoning skills (object), the community context is defined by heterogeneity, where student capabilities within a single class often span several grades. This disparity creates a conflict in the division of labor: a single teacher cannot simultaneously provide the individualized attention required to keep every student within their optimal Zone of Proximal Development (ZPD) [10]. Consequently, students often turn to the alternative avenues, such as standard AI chat-bots, to procure answers, rather than engaging in problem-solving.

As illustrated in Figure 1, the primary systemic contradiction emerges between the division of labor (a single teacher) and the heterogeneous classroom community. This misalignment results in insufficient individualized activity within students' ZPD.

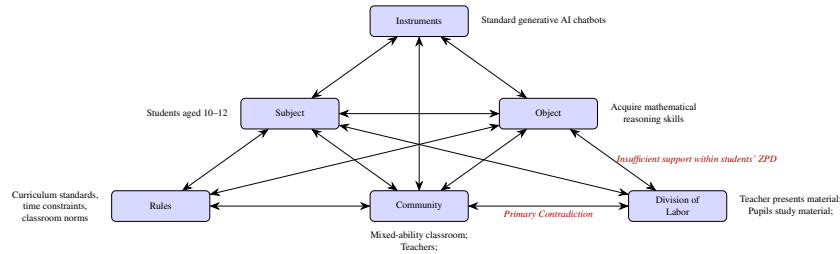


Figure 1. Engeström's activity system of the current mathematics classroom.

4.2. Use Scenario

The user, a learner, plays a card-based combat game as illustrated in Figure 2 where each card is locked by a mathematical challenge. To unlock and play a card, the learner must solve the associated problem; incorrect answers result in a reduction of card power. The goal is to defeat an adversary and advance to a further stage. When stuck on a challenge, the learner can prompt an in-game agent for help. The agent does not provide the answer directly, but instead asks guiding questions to help the learner reason through the problem. Over the course of a session, the system adapts to the learner's strengths and weaknesses, tailoring both the difficulty of challenges and the guidance provided.

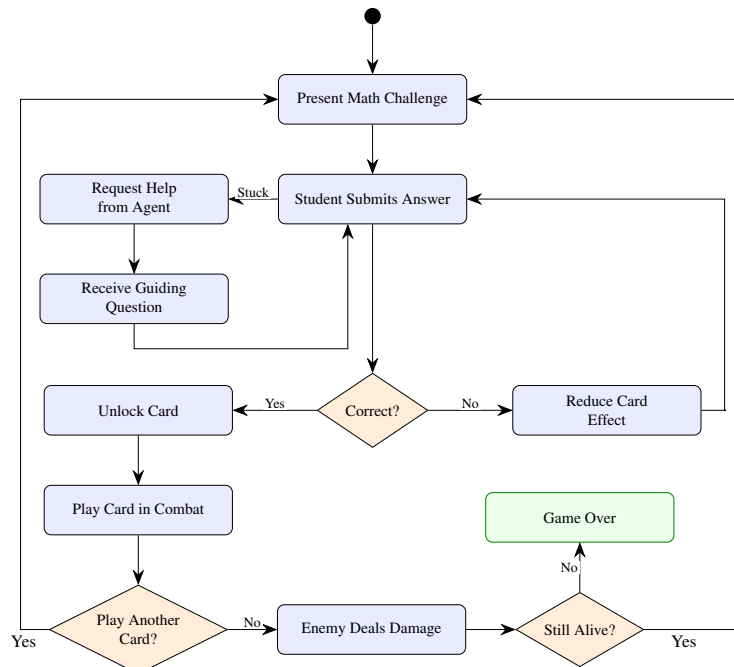


Figure 2. Activity diagram of the gameplay loop.

4.3. Procedure for the Participatory Design Process Involving Stakeholders

To ensure the system aligns with both pedagogical standards and the motivational needs of the target group, the design procedure organizes collaboration with a mathematics teacher and students aged 10–12. We conduct periodic check-ins with a teacher to define goals, mechanics, and inclusion needs, and final prototype evaluation with an undetermined amount of learners. During the evaluation, students are asked to critique the dialogue style, thereby allowing them to shape the “rewards” that support their sense of autonomy and competence [8]. Each meeting produces prioritized action items to directly inform the next iteration.

4.4. Anticipated Design Solution that will be Evaluated with Tentative Users

The core of the design is a dynamic User Model operationalized within a RAG architecture, designed to function as a “More Knowledgeable Peer”. The system employs the “MiniGuide” strategy as a strict constraint on the RAG generation layer, thereby forcing the LLM to generate guiding questions that maintain the learner within their ZPD. Figure 3 shows how these components interact at the system level.

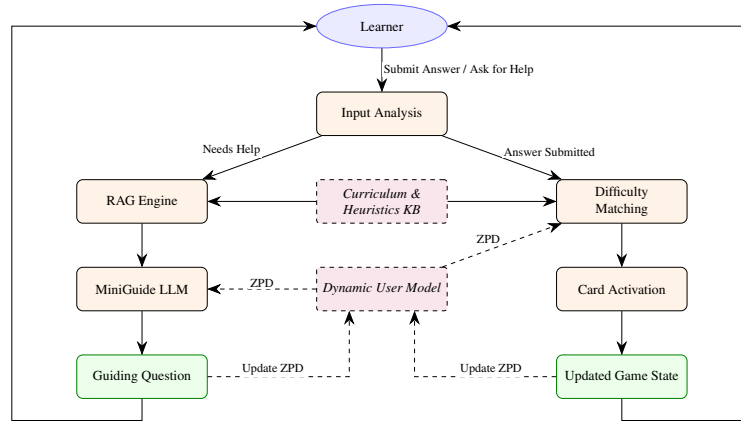


Figure 3. Data flow diagram of the system architecture. Solid arrows indicate the primary data flow; dashed arrows indicate reads from and writes to the Dynamic User Model.

4.4.1. Game Design

The learner plays a turn-based card combat game inspired by the roguelike deck-building genre. The player is presented with a hand of cards, each displaying its power value and associated challenge difficulty (Figure 4). Each card in the player’s hand is locked behind a mathematical challenge matched to their current ability level. To play a card, the learner must correctly solve the associated problem. The goal is to defeat an adversary and advance to the next stage, creating a gameplay loop where mathematical reasoning is the core mechanic rather than an interruption.

Cards vary in power, and stronger cards are gated by harder problems. This coupling serves a dual purpose: it motivates learners to attempt more difficult challenges for strategic advantage, and it naturally prevents progress without genuine mathematical



Figure 4. The battle screen after a single turn. In the top right we see the players health and active shield. Centered on the screen is the enemy display, showing their health and damage that will be dealt next turn. Cards and their type, strength, energy cost, as well as topic and difficulty of the associated math challenge are displayed in the bottom.

engagement. Crucially, incorrect answers carry an in-game consequence in the form of card power reduction, mirroring the stakes of an actual game combat encounter. This mechanic discourages random guessing, ensuring that learners engage thoughtfully with each problem rather than treating the math as a low-cost obstacle to click through [14].

4.4.2. Scaffolding Through the “MiniGuide” Strategy

When a learner is stuck on a challenge, they can request help from an in-game agent (Figure 5). Rather than providing the answer, the agent follows the “MiniGuide” strategy: it first diagnoses the learner’s current state using the User Model, then retrieves relevant curricular content from a knowledge base via RAG, and finally generates a guiding question designed to nudge the learner toward the solution. This preserves the learner’s agency and keeps them actively reasoning.

The strict constraint on the generation layer aims to prevent the LLM from producing direct answers. Instead, its output is limited to diagnostic questions, hints, and prompts for reflection — functioning as a peer who helps the learner think, not one who thinks for them.

4.4.3. Dynamic User Model

Throughout a learning session, the system builds and refines a User Model that captures the learner’s current mastery across mathematical topics, common error patterns, and help-seeking behavior. This model informs both the difficulty of challenges presented and the type of guidance the agent provides. For example, a learner who consistently struggles with fractions will receive challenges and guidance tailored to that gap, while a learner who frequently requests hints without attempting a solution first may receive prompts that encourage independent effort before offering guidance.

In practice, the system tracks correct and incorrect answer attempts and hint requests per difficulty, aggregated by topic. The content of the user model is presented in the user interface as the “Elevprofil” (Figure 6). When generating new tasks or guiding questions, these metrics are summarized using simple threshold based rules into a learner profile, as shown in Figure 7, identifying strengths, weaknesses, and behavioral patterns, which

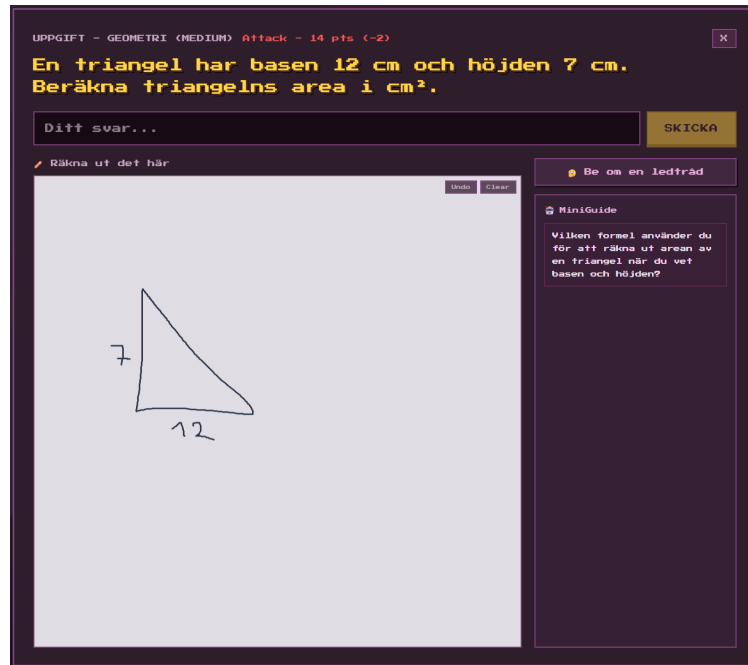


Figure 5. The task screen for an easy “Tid & Datum” math task, where the answer has been submitted, and the user has requested a hint. The player is presented with the question, a field for entering the answer, a whiteboard for sketching out solutions, and a button for getting a hint with the list of past hints below.

is then included in the RAG context for subsequent interactions. This means the agent’s responses become increasingly personalized over the course of a session, allowing the system to adapt both game difficulty and pedagogical approach to keep the learner within their ZPD.

ELEVPROFIL			
ÅTERSTÄLL			
ÄMNE	FÖRSÖK	RÄTT	LEDTRÄDAR
Enheter	3	2	2
EASY 1/1 MEDIUM 1/2			
Geometri	20	8	1
EASY 5/14 MEDIUM 3/6			
Statistik	2	2	0
EASY 1/1 MEDIUM 1/1			
De fyra räknesätten	2	2	0
MEDIUM 1/1 HARD 1/1			

Figure 6. The user model (here called the student profile). The number of answer attempts, correct answers and number of hints received are displayed, grouped by topic.

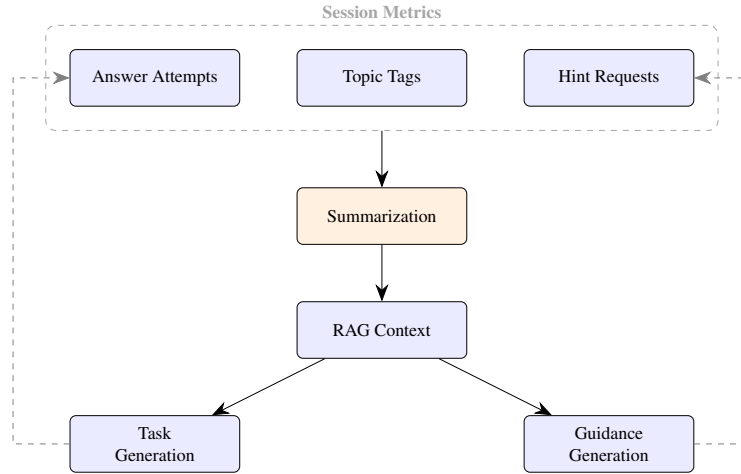


Figure 7. User Model creation process.

4.4.4. Determining the Zone of Proximal Development

The system approximates each learner's Zone of Proximal Development by classifying problem interactions into three categories based on the level of assistance required.

Below: Problems that the learner solves correctly without requesting any hints, these represent mastered material.

[$Accuracy \geq 70\%$ & $hints = 0$]

Above: Problems where the learner fails to arrive at the correct answer despite available assistance, these are beyond the learner's current reach.

[$Accuracy < 40\%$ & $hints \geq 1$]

Within: Problems that the learner solves correctly after receiving one or more guiding questions from the agent, these are the problems where scaffolding is most effective and where learning is most likely to occur.

This classification, summarized in Figure 8, is tracked per topic and accumulated over a session in the form of the user model.

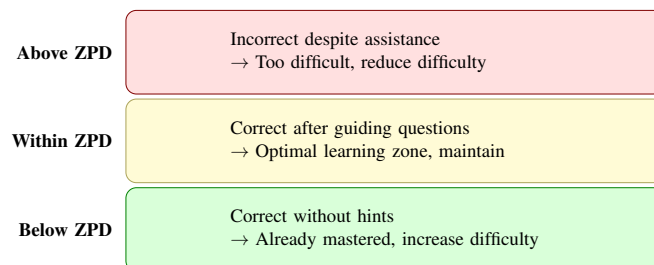


Figure 8. Classification of problem interactions into ZPD regions.

4.4.5. Gamification and Motivation

The game structure is designed to align with Self-Determination Theory [8] by supporting autonomy (choosing which cards to attempt), competence (solving challenges at an appropriate difficulty), and relatedness (interacting with the in-game agent as a supportive peer). Because card power is tied to problem difficulty relative to the learner’s level, both high-performing and struggling learners face meaningful challenges — avoiding boredom for advanced students and reducing frustration for those who need more support.

Importantly, progress in the game is closely tied to demonstrated mathematical reasoning, and not surpassable game mechanics or repetition. This transforms the friction of learning into a core part of the game loop rather than an obstacle to it.

4.5. Material and software

The prototype adopts a decoupled client-server web architecture to enable rapid prototyping and broad accessibility on standard school devices without requiring software installation. The frontend is developed using Next.js, a modern web framework built upon React, to manage the interactive gamified interface. Server-side logic is handled through a Python FastAPI backend, which orchestrates the RAG pipeline, game logic, and user modeling components.

A multimodal Large Language Model is required to process handwritten mathematical input from the canvas interface. The prototype uses Anthropic’s Claude Sonnet 4.6 for both handwriting recognition and guided response generation, but the long-term architecture anticipates replacing the third party model with a self-hosted, or EU-based model to better align with data privacy requirements for use in schools.

The RAG framework is connected to a curriculum-aligned knowledge base indexed within a vector database. For mathematical content, the system draws on Matteboken.se, a free and comprehensive digital mathematics resource provided by the Swedish non-profit Mattecentrum. The platform covers the full Swedish curriculum for mellanstadiet (grades 4–6), including arithmetic operations, geometry, units of measurement, statistics, and time, providing a structured and openly accessible content base for the retrieval corpus.

4.5.1. LLM prompting and output stability

The system communicates with the LLM using three system prompts, each strictly scoped to a single responsibility: task generation, guidance generation, and topic & difficulty selection. All prompts follow the same structure: a role definition, explicit behavioral constraints, and a required JSON output schema.

To ensure stable and parseable output, all prompts require the LLM to respond with a JSON object matching a predefined schema, with no additional text, markdown formatting, or preamble. This allows the backend to parse responses without relying on text extraction. When the response does not conform to the expected schema, the system rejects it and retries the request rather than passing malformed output to the student.

4.6. Evaluation setup

The evaluation adopts an exploratory, qualitative approach. It consists of two sessions with distinct participant groups.

The first session involves a teacher acting as the domain expert during which we focus on evaluating the RAG system’s generation layer: whether the guiding questions produced by the “MiniGuide” strategy are pedagogically sound, and question content and difficulty is appropriate for the target grade. The teacher interacts with the system and provides feedback on the state of the prototype.

During the second session, the system is distributed to an undetermined number of school students aged 10–12 (2-3 classrooms, upward of 20 students). This session focuses on usability and engagement rather than learning outcomes, as a single session is insufficient to measure knowledge gains. Students are presented with a simple questionnaire to capture the students’ disposition towards the system. The questionnaire is built into the system (Figure 9) and consists of the following questions:

- **Q1:** How would you rate your math skill level? (1–5)
“Hur bedömer du din matematikförmåga?”
- **Q2:** How difficult do you find the math tasks? (1–5)
“Hur svåra tycker du att matematikuppgifterna är?”
- **Q3:** How helpful are the hints when you use them? (1–5)
“Hur hjälpsamma är ledtrådarna när du använder dem?”
- **Q4:** How likely are you to play this in your spare time? (1–5)
“Hur troligt är det att du spelar det här på fritiden?”
- **Q5:** How stressed do you feel while playing? (1–5)
“Hur stressad känner du dig när du spelar?”
- Gender (Male, Female, Other, Prefer not to say)
- Age

5. Ethical Considerations

The involvement of children aged 10–12 in evaluation sessions requires informed consent from both parents and participants, with clear communication that participation is voluntary and withdrawal is possible at any time. During the student demo session, no personally identifiable information will be collected, and any observational notes will be anonymized.

The prototype currently relies on Anthropic’s Claude, meaning student input — including handwritten mathematical work — is processed by a third-party cloud service outside the EU. For the evaluation phase, this is managed by ensuring no personal data is included in the prompts. This dependency motivates the planned shift toward a locally hosted or EU-based model, better aligning with GDPR requirements for use in Swedish schools.

Finally, positioning an AI as a “More Knowledgeable Peer” carries a risk of displacing peer collaboration and teacher interaction over time. The “MiniGuide” strategy mitigates passive dependency by never providing direct answers, but in a long-term deployment the system should remain a complement to human instruction, with teachers retaining full pedagogical authority [15].

ÅTERKOPPLINGSENKÄT ✕

Betygsätt varje fråga från 1 till 5.

Hur bedömer du din matematikförmåga?

1 2 3 4 5

Hur svåra tycker du att matematikuppgifterna är?

1 2 3 4 5

Hur hjälpsamma är ledtrådarna när du använder dem?

1 2 3 4 5

Hur troligt är det att du spelar det här på fritiden?

1 2 3 4 5

Hur stressad känner du dig när du spelar?

1 2 3 4 5

Vilket kön har du?

Man Kvinna Annat Vill ej uppge

Hur gammal är du?

0/7 besvarade SKICKA

Figure 9. Questionnaire integrated into the system.

6. Results

The outcome is a fully functional prototype¹ that demonstrates the core interaction loop: solving math challenges to unlock cards, receiving guided scaffolding when stuck, and progressing through combat encounters. The prototype includes complementary features such as a CI/CD pipeline for automated deployment and user persistence via Google account login, allowing students to retain their profiles across sessions.

The teacher confirmed that the guiding questions produced by the MiniGuide strategy were pedagogically appropriate for the target age group. The generated math tasks were judged to match the content and difficulty present in the standard curriculum.

From the student demo session, unfortunately we did not receive sufficient survey responses to derive informative conclusions, but saw sufficient engagement with the system to draw insights from.

¹<https://github.com/jleipus/math-quest>

7. Discussion

The participatory design methodology provided a useful structure for prototyping while remaining inline with pedagogical needs, but its application was constrained by the short timeline and limited stakeholder access. In practice, the iterative cycle of co-design, prototyping, and reflection was reduced to a single round of teacher feedback followed by a student demo session, rather than the multiple iterations the methodology calls for.

The interaction log data serves as the primary basis for analysis, as the low survey response rate prevents drawing meaningful conclusions from the questionnaire alone. In hindsight, the survey should have been incentivized — for example, by integrating it as a required step before a final game round or by offering an in-game reward for completion.

7.1. Addressing the Research Questions

RQ1: How can a dynamic user model approximate a student's ZPD and deliver guidance that promotes active reasoning?

Figure 10 shows that accuracy decreases and hint usage increases as difficulty rises from easy to hard, suggesting that easy problems mostly fall below the ZPD, while hard problems generally lie above it. Medium difficulty tasks proved to be closest to the targeted ZPD. However, the system currently treats all errors equally – a calculation error is penalized the same way as a conceptual error, which limits the precision of the ZPD approximation.

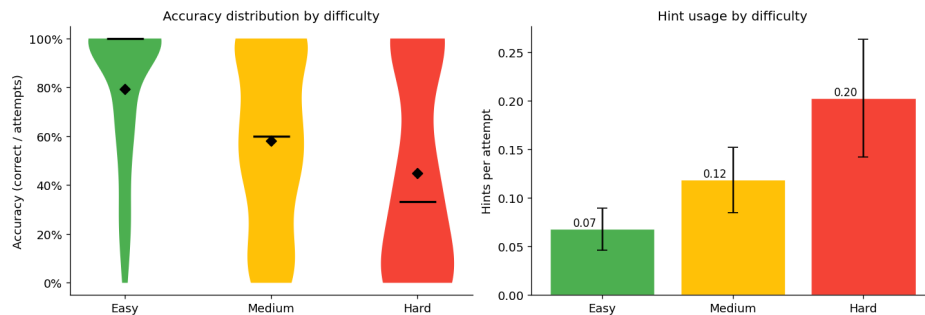


Figure 10. Left: violin plot of answer accuracy distribution by difficulty. Line shows median value, diamond shows mean value. Right: bar chart of average hint usage by difficulty.

RQ2: To what extent can game-based design reframe errors as learning moments and sustain motivation?

Figure 11 shows most users attempted fewer than 10 problems, while a small group engaged extensively. Survey responses on stress (Q5), while highly limited, were generally low.

The evaluation does not address what happens when a student answers incorrectly multiple times in a row. Repeated failure combined with the card power reduction mechanic could become discouraging, particularly for younger or less confident learners.

RQ3: How can the MiniGuide strategy ensure that AI responses provide guidance rather than direct answers?

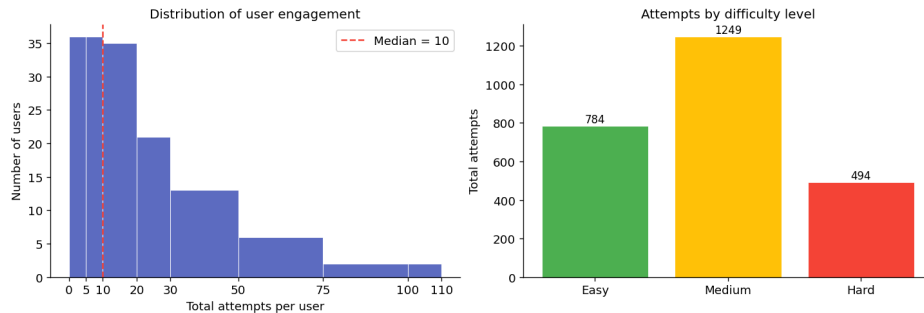


Figure 11. Left: histogram of total answer attempts made by a user. Right: bar chart of overall number of answer attempts by question difficulty.

Evaluation with the teacher confirmed that the guidance prompt responses were pedagogically sound and avoided leaking answers, but further evaluation and additional survey responses are required to more accurately gauge the system’s success at addressing this question.

7.2. Limitations

The evaluation is limited in scope. Two sessions is insufficient for measuring learning outcomes, long-term engagement, or the effectiveness of the adaptive difficulty system. This would require longitudinal studies with larger and more diverse participant groups, including students across different ability levels.

The current prototype does not implement content guardrails to filter potentially inappropriate LLM output before it reaches the student. While no issues were observed during testing, this is a necessary safeguard before any unsupervised use with children. This would need to be addressed in future iterations.

7.3. Self-Categorization Theory

Another theory that was considered when building the theoretical grounding for the project was Self-Categorization Theory (SCT), which describes how individuals classify themselves and others into social groups, and how these categorizations influence behavior and self-perception. In an educational context, this is relevant because students often categorize themselves as “good at math” or “bad at math,” which can affect their willingness to engage with challenging material. The current system does not account for these self-categorizations – it adapts to observable performance rather than the student’s identity or group-based beliefs about their ability. Investigating how the system interacts with these identity dynamics would be a valuable direction for future work.

8. Conclusions and Future Work

Our aim for this project was to investigate how Active Learning in primary mathematics education can be systematically supported through a responsible, interactive AI system. We successfully implemented a RAG-based gamefied system that provides adaptive scaf-

folding, and attempted to evaluate it on the target demographic. While we were unable to collect sufficient data to properly answer all the research questions we had raised, strong positive indications were observed.

Further work with this system would explore options for expanding the content base used as RAG context. Additionally, the system could be adapted to support a wider variety of question formats (e.g. images, non-numeric answers). With these improvements, it would be interesting to see whether the system could be expanded to target not only 4–6 grades, but elder students as well.

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